CURRENT ONE-STAGE AND TWO-STAGE IPDE DESIGN WITH EFFICACY

Updated carryover-model options and time-to-event implementation

Purpose. This document updates the efficacy-enabled AIDE/IPDE framework to include two toxicity carryover models, four efficacy carryover models, and a time-to-event implementation that combines a fixed-r TITE-CRM likelihood for toxicity with a delayed-outcome likelihood approximation for efficacy.

Design family	AIDE/IPDE with binary toxicity and binary efficacy
Dose levels	$j = 1, \dots, J$
Trial structures	One-stage and two-stage
Toxicity options	Shared r_T complement-discount model; shared α_T additive previous-dose model
Efficacy options	r_E or α_E, each specified either dose-specifically or shared across dose levels
TITE implementation	Toxicity: fixed-r_T TITE-CRM; efficacy: delayed-outcome approximation
Enrollment schemes	Continuous enrollment with new-patient priority, or IPDE-first recruitment
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Technical design specification. The delayed-outcome efficacy component is an endpoint adaptation of the likelihood approximation developed for delayed binary outcomes; it must be evaluated through simulation for the intended efficacy assessment window and response-time distribution.

Design Summary

Core principle. Toxicity determines safety, admissibility, and the allowable direction of dose movement. Efficacy or utility ranks doses only after toxicity restrictions have been applied. An IPDE treatment is allowed only when its individual toxicity-risk criterion and expected efficacy-improvement criterion are both satisfied.

The trial structure, enrollment mechanism, carryover model, and delayed-outcome implementation are separate design choices except that the current TITE toxicity implementation is explicitly restricted to the fixed-r_T model.

Model option matrix

Endpoint	Option	Parameter scope	IPDE probability
Toxicity	r_T complement-discount	Shared across levels	$q_{T,i} = r_T + (1 - r_T)p_{T,j_i}$
Toxicity	α_T additive previous-dose	Shared across levels	$q_{T,i} = \min\{1, p_{T,j_i} + \alpha_T p_{T,\ell_i}\}$
Efficacy	$r_{E,j}$ complement-discount	Dose-specific	$q_{E,i} = r_{E,j_i} + (1 - r_{E,j_i})p_{E,j_i}$
Efficacy	$\alpha_{E,j}$ additive previous-dose	Dose-specific	$q_{E,i} = \min\{1, p_{E,j_i} + \alpha_{E,j_i} p_{E,\ell_i}\}$
Efficacy	r_E complement-discount	Shared across levels	$q_{E,i} = r_E + (1 - r_E)p_{E,j_i}$
Efficacy	α_E additive previous-dose	Shared across levels	$q_{E,i} = \min\{1, p_{E,j_i} + \alpha_E p_{E,\ell_i}\}$

Fixed versus estimated carryover. A shared or dose-specific carryover coefficient may be fixed at a prespecified value or estimated with a prespecified Beta prior. The model definitions below show the estimated version unless explicitly labeled fixed. In the current TITE toxicity implementation, $r_T = r_T^{\text{fix}}$ is supplied as data and is not sampled.

Notation

Symbol	Definition	Symbol	Definition
i	Administration-level index	j	Generic dose-level index
j_i	Current dose for administration i	ℓ_i	Same patient's immediately preceding dose
I_i	1 for IPDE/recycled administration; 0 for regular	$p_{T,j}$	Regular-patient toxicity probability at dose j
$p_{E,j}$	Regular-patient efficacy probability at dose j	$q_{T,i}$	Model-specific IPDE toxicity probability
$q_{E,i}$	Model-specific IPDE efficacy probability	ϕ_T	Target toxicity probability
ϕ_E	Minimum acceptable efficacy for futility	c_T, c_E	Posterior monitoring cutoffs
δ	Minimum expected efficacy improvement for IPDE	H	Highest dose level tried at least once
$u_{T,i}, u_{E,i}$	Follow-up accumulated by interim time	τ_T, τ_E	Toxicity and efficacy assessment windows
$w_{T,i}, w_{E,i}$	Endpoint-specific follow-up weights	$\Delta_{E,i}$	Efficacy ascertainment indicator

1. Statistical Model

1.1 Baseline toxicity CRM

Let p_{0j} be the prespecified CRM skeleton probability at dose j, and let β be the one-dimensional CRM parameter. Under a power CRM,

$$p_{T,j}(\beta) = p_{0j}^{\exp(\beta)}, \qquad \beta \sim N(\mu_\beta, \sigma_\beta^2). \tag{1}$$

For a regular administration at dose j_i,

$$Y_{T,i} \mid \beta, I_i = 0 \sim \text{Bernoulli}\{p_{T,j_i}(\beta)\}. \tag{2}$$

The IPDE probability is determined by one of the following two toxicity options.

1.2 Toxicity option T-R: shared complement-discount parameter r_T

For an IPDE administration at current dose j_i,

$$q_{T,i}^{(r)} = r_T + (1 - r_T)p_{T,j_i}(\beta), \qquad 0 \leq r_T \leq 1. \tag{3}$$

When estimated, $r_T \sim \text{Beta}(a_{r_T}, b_{r_T})$. When fixed, $r_T = r_T^{\text{fix}}$ is user-specified. The parameter is shared across dose levels, so all IPDE toxicity observations inform the same coefficient. When $r_T = 0$, the IPDE and regular probabilities are identical; increasing r_T shifts the IPDE risk toward one.

1.3 Toxicity option T-A: shared additive previous-dose parameter α_T

For an IPDE administration at current dose j_i after a preceding dose ℓ_i,

$$q_{T,i}^{(\alpha)} = \min\left\{1,\ p_{T,j_i}(\beta) + \alpha_T p_{T,\ell_i}(\beta)\right\}, \qquad \alpha_T \sim \text{Beta}(a_{\alpha_T}, b_{\alpha_T}). \tag{4}$$

The coefficient α_T is shared across IPDE administrations, while the added risk depends on the patient's actual preceding dose. The cap at one is applied administration by administration.

Current TITE restriction. Both T-R and T-A are complete-outcome toxicity model options. The current time-to-event specification in Section 2 uses T-R with a fixed r_T only. A TITE implementation of T-A would require a separate software extension and is not part of the present design specification.

1.4 Baseline efficacy model

The regular efficacy probability is modeled separately at each dose:

$$p_{E,j} \sim \text{Beta}(a_{E,j}, b_{E,j}), \qquad j = 1, \ldots, J. \tag{5}$$

Equivalently, the individual-level model is

$$Y_{E,i} \mid p_{E,j_i}, I_i = 0 \sim \text{Bernoulli}(p_{E,j_i}). \tag{6}$$

After aggregation of regular administrations at dose j, this produces the usual beta-binomial updating. Individual rows are retained in implementation because the additive IPDE options depend on the preceding dose.

1.5 Four efficacy carryover options

1.5.1 E-RD: r model with dose-specific carryover

$$r_{E,j} \sim \text{Beta}(a_{r,j}, b_{r,j}), \tag{7}$$

$$q_{E,i}^{(r,\text{dose})} = r_{E,j_i} + (1 - r_{E,j_i})p_{E,j_i}. \tag{8}$$

Each destination dose has its own $r_{E,j}$. IPDE data at dose j directly inform that dose's carryover coefficient.

1.5.2 E-AD: α model with dose-specific carryover

$$\alpha_{E,j} \sim \mathrm{Beta}(a_{\alpha,j}, b_{\alpha,j}), \tag{9}$$
$$q_{E,i}^{(\alpha,\mathrm{dose})} = \min\left\{1,\ p_{E,j_i} + \alpha_{E,j_i} p_{E,\ell_i}\right\}. \tag{10}$$

The additive coefficient is indexed by the destination dose j_i, while the amount added depends on the preceding-dose efficacy probability p_{E,ℓ_i}.

1.5.3 E-RS: r model shared across dose levels

$$r_E \sim \mathrm{Beta}(a_r, b_r), \tag{11}$$
$$q_{E,i}^{(r,\mathrm{shared})} = r_E + (1 - r_E) p_{E,j_i}. \tag{12}$$

One shared r_E pools IPDE information across all destination doses.

1.5.4 E-AS: α model shared across dose levels

$$\alpha_E \sim \mathrm{Beta}(a_\alpha, b_\alpha), \tag{13}$$
$$q_{E,i}^{(\alpha,\mathrm{shared})} = \min\left\{1,\ p_{E,j_i} + \alpha_E p_{E,\ell_i}\right\}. \tag{14}$$

This is the shared-parameter additive previous-dose model. Two IPDE administrations at the same destination can have different probabilities if their preceding doses differ.

1.6 Gamma-ratio representation of Beta priors

For JAGS implementations in which the direct Beta density is to be represented through Gamma variables, any parameter $c \sim \mathrm{Beta}(a, b)$ may be written as

$$g_1 \sim \mathrm{Gamma}(a, 1), \qquad g_2 \sim \mathrm{Gamma}(b, 1), \qquad c = \frac{g_1}{g_1 + g_2}. \tag{15}$$

The Gamma variables must use a common rate. This representation applies to $p_{E,j}$ and to every shared or dose-specific r or α coefficient. It is mathematically equivalent to the intended Beta prior.

1.7 Complete-outcome likelihood

Let

$$\theta_{T,i} = (1 - I_i) p_{T,j_i} + I_i q_{T,i}, \qquad \theta_{E,i} = (1 - I_i) p_{E,j_i} + I_i q_{E,i}, \tag{16}$$

where $q_{T,i}$ and $q_{E,i}$ are selected from the active model options. With complete binary outcomes,

$$L_T \propto \prod_{i=1}^{N_T} \theta_{T,i}^{Y_{T,i}} (1 - \theta_{T,i})^{1 - Y_{T,i}}, \tag{17}$$
$$L_E \propto \prod_{i=1}^{N_E} \theta_{E,i}^{Y_{E,i}} (1 - \theta_{E,i})^{1 - Y_{E,i}}. \tag{18}$$

For complement-discount models, rows can be grouped by current dose and administration type. For additive previous-dose models, the individual-level form is required because the pair (ℓ_i, j_i) changes the IPDE probability.

Decision summary convention. Posterior means are the default point estimates in the operating rules. Posterior medians may be substituted, but the same convention must be used across all doses, endpoints, and decision times.

2. Time-to-Event and Delayed-Outcome Implementation

2.1 Interim endpoint notation

At an interim decision time, each administration has an assigned dose, administration type, observed event indicator by that time, and accumulated follow-up. Endpoint-specific weights are clipped to $[0, 1]$. Under the default uniform event-time assumption,

$$w_{T,i} = \min\left\{1, \max\left(0, \frac{u_{T,i}}{\tau_T}\right)\right\}, \qquad w_{E,i} = \min\left\{1, \max\left(0, \frac{u_{E,i}}{\tau_E}\right)\right\}. \tag{19}$$

Observed toxicity events, observed efficacy responses, and fully completed assessment windows are treated as ascertained outcomes. The toxicity and efficacy endpoints use different likelihood conventions below.

2.2 Toxicity: fixed-r_T TITE-CRM likelihood

The current TITE toxicity model uses the shared complement-discount model with a user-specified constant r_T^{fix}. Define the pre-TITE row probability

$$\theta_{T,i} = (1 - I_i)p_{T,j_i}(\beta) + I_i\left[r_T^{\text{fix}} + (1 - r_T^{\text{fix}})p_{T,j_i}(\beta)\right]. \tag{20}$$

For a DLT, $w_{T,i}$ is forced to one. The TITE-weighted event probability is

$$\theta_{T,i}^{\text{eff}} = w_{T,i}\theta_{T,i}, \tag{21}$$

and the interim likelihood is

$$L_T^{\text{TITE}}(\beta \mid r_T^{\text{fix}}) = \prod_{i=1}^{N_T}(w_{T,i}\theta_{T,i})^{Y_{T,i}}(1 - w_{T,i}\theta_{T,i})^{1-Y_{T,i}}. \tag{22}$$

Toxicity status	Convention	Likelihood contribution
Observed DLT	$Y_{T,i} = 1$, $w_{T,i} = 1$	$\theta_{T,i}$
Completed non-DLT	$Y_{T,i} = 0$, $w_{T,i} = 1$	$1 - \theta_{T,i}$
Pending non-DLT	$Y_{T,i} = 0$, $0 < w_{T,i} < 1$	$1 - w_{T,i}\theta_{T,i}$
No follow-up	$Y_{T,i} = 0$, $w_{T,i} = 0$	1

Toxicity convention. The pending non-DLT contribution is $1 - w_{T,i}\theta_{T,i}$. The delayed-outcome approximation $(1 - \theta_{T,i})^{w_{T,i}}$ is not used for toxicity in this design. When $r_T^{\text{fix}} = 0$, the model reduces to the original TITE-CRM likelihood for regular and IPDE rows.

2.3 Efficacy: delayed-outcome likelihood approximation

Let $\theta_{E,i}$ be the row-specific efficacy probability generated by one of E-RD, E-AD, E-RS, or E-AS. Let $\Delta_{E,i} = 1$ when the efficacy endpoint is ascertained by the interim time, meaning that a response has occurred or the full efficacy window is complete; otherwise $\Delta_{E,i} = 0$ and the patient is pending without an observed response.

For a pending nonresponse, the exact event-time observed-data contribution under weight $w_{E,i} = \Pr(T_{E,i} \le u_{E,i} \mid Y_{E,i} = 1)$ is $1 - w_{E,i}\theta_{E,i}$. The delayed-outcome approach replaces that term by the first-order approximation

$$1 - w_{E,i}\theta_{E,i} \approx (1 - \theta_{E,i})^{w_{E,i}}. \tag{23}$$

The resulting efficacy likelihood is

$$L_E^{\text{DO}} \propto \prod_{i=1}^{N_E}\theta_{E,i}^{\Delta_{E,i}Y_{E,i}}(1 - \theta_{E,i})^{\Delta_{E,i}(1-Y_{E,i})+(1-\Delta_{E,i})w_{E,i}}. \tag{24}$$

Thus, an observed response contributes $\theta_{E,i}$, a completed nonresponse contributes $1 - \theta_{E,i}$, and a pending nonresponse contributes $(1 - \theta_{E,i})^{w_{E,i}}$.

Efficacy status	Convention	Likelihood contribution
Observed response	$Y_{E,i} = 1$, $\Delta_{E,i} = 1$	$\theta_{E,i}$
Completed nonresponse	$Y_{E,i} = 0$, $\Delta_{E,i} = 1$	$1 - \theta_{E,i}$
Pending nonresponse	$Y_{E,i} = 0$, $\Delta_{E,i} = 0$	$(1 - \theta_{E,i})^{w_{E,i}}$
No efficacy follow-up	$w_{E,i} = 0$	1

Endpoint adaptation and validation. The cited delayed-outcome derivation was developed and evaluated for delayed toxicity. The present design applies the same binary-event approximation to efficacy response. This extension should be validated by simulation under plausible efficacy response-time distributions, assessment-window lengths, accrual rates, and model misspecification.

2.4 Combined interim analysis

At each model update:

1. Fit the fixed-r_T TITE-CRM toxicity model using all started toxicity rows and their $w_{T,i}$ values.
2. Fit the selected efficacy carryover model using ascertained efficacy outcomes and weighted pending nonresponses through L_E^{DO}.
3. Use posterior summaries of $p_{T,j}$ for dose-level safety and MTD decisions, and model-specific $q_{T,i}$ for individual IPDE safety.
4. Use posterior summaries of $p_{E,j}$ for efficacy ranking, futility, utilities, and final OBD selection, and model-specific $q_{E,i}$ for IPDE improvement.

One-stage decision interface. The one-stage design does not use a separate m_U threshold. At each decision, the selected toxicity and efficacy models define the current admissible set and utility values. The utility-maximizing admissible dose is the provisional OBD; if that dose lies above the highest tried dose, upward movement follows the no-skipping rule in Section 3.2.

3. IPDE Trial Design

At every decision time, safety is evaluated before efficacy or utility. Doses eliminated for overtoxicity or efficacy futility are removed from cohort-allocation and IPDE-destination sets.

3.1 Two-stage design

1. Stage I: toxicity-driven AIDE. Treat patients according to the toxicity-based AIDE design until the cumulative Stage I sample size reaches *s1_Max* and the next toxicity recommendation is *stay*. Efficacy outcomes are collected and monitored, but Stage I cohort assignments are determined by toxicity.

2. Stage II admissible set. After Stage I and after every subsequent Stage II cohort, update the toxicity and efficacy models, recalculate the current MTD, and reassess overtoxicity and futility. Define A_{II} as the doses at or below the current MTD that are not eliminated for overtoxicity or futility. No tried-dose restriction is imposed on Stage II allocation.

3. Observed efficacy-response indicator. For each dose j, let R_j=1 if dose j has produced at least one observed efficacy response and R_j=0 otherwise. This indicator modifies Stage II allocation but does not replace the model-based utility or futility calculations.

4. Stage II version B: highest-utility allocation. Let A_R={j in A_{II}: R_j=1}. If A_R is nonempty, assign the entire next cohort to the dose in A_R with the largest utility. If no admissible dose has an observed efficacy response, assign the cohort to the current MTD.

5. Stage II version A: top-two utility randomization. First identify the ordinary two highest-utility doses in A_{II}. If both have at least one efficacy response, randomize between them. If exactly one has a response, randomize between that responder dose and the current MTD. If neither has a response, randomize between the current MTD and the highest admissible dose below the MTD. If the rule produces a duplicate dose, use the highest distinct admissible alternative below the MTD; if only one admissible dose remains, assign the full cohort to that dose. Randomization probabilities remain proportional to the prespecified nonnegative utility.

6. Stage II updating. After every evaluated Stage II cohort, repeat the complete model update, MTD calculation, overtoxicity and futility assessment, response-indicator update, admissible-set construction, and Stage II allocation rule.

> **Stage II response-informed safeguard.** When no admissible dose has demonstrated efficacy, the design does not automatically favor a lower-toxicity dose. Instead, the MTD (or the MTD and the highest admissible dose below it for top-two randomization) receives additional opportunity to demonstrate efficacy, subject to continued toxicity and futility monitoring.

3.2 One-stage design

1. Update the admissible set. Refit the prespecified toxicity and efficacy models, recalculate the model-based MTD, overtoxicity elimination, and efficacy futility, and define the current allocation-admissible set.

2. Compute the provisional OBD. Compute the prespecified utility U(j) for every allocation-admissible dose and let j_U be the admissible dose with the largest utility, using the prespecified tie-breaking rule.

3. Escalation and stay decisions. Let d be the current dose and H the highest dose tried at least once. If j_U>d, proceed toward j_U using the existing no-skipping escalation rule; if j =d, remain at d.

4. Response-informed downward allocation. If j_U<d, define A_{down}={j: j_U<=j<=d, j is toxicity-admissible and not otherwise eliminated, and R_j=1}. If A_{down} is nonempty, assign the next cohort to min(A_{down}), the lowest response-qualified admissible dose between the utility target j_U and the current dose d.

5. No response-qualified downward destination. If A_{down} is empty and the current dose remains toxicity-admissible, remain at the current dose. If the current dose is no longer toxicity-admissible, safety overrides efficacy and the next cohort is assigned to the highest toxicity-admissible dose below the current dose, subject to the global elimination rules.

6. Repeat after each cohort. After the cohort becomes evaluable, update the models, MTD, elimination status, response indicators, utilities, and allocation decision again.

3.3 Enrollment schemes

Scheme	Queue priority	Operational rule
Continuous enrollment	New patients first	Enrollment remains open; waiting new patients fill cohort positions before eligible IPDE patients.
IPDE-first recruitment	Eligible IPDE patients first	Eligible IPDE patients fill first; recruit only enough new patients to complete the cohort.

Event-clock operation. New-patient arrivals and completed-cycle IPDE decisions are assignable events. If no cohort position is open, the corresponding opportunity is placed in the waiting queue. The selected enrollment scheme determines queue priority; the one-stage or two-stage design determines the assigned dose.

4. Overtoxicity, Futility, and IPDE Criteria

The following checks are repeated whenever outcomes or follow-up information change the posterior distributions.

4.1 Overtoxicity monitoring

For each dose j, eliminate the dose for overtoxicity when

$$\Pr(p_{T,j} > \phi_T \mid \mathcal{D}) > c_T. \tag{26}$$

An eliminated dose cannot receive a new cohort or serve as an IPDE destination. Under toxicity monotonicity, elimination of dose j may also eliminate all higher doses. The trial terminates early when no admissible safe dose remains, including when the lowest dose is eliminated.

4.2 Efficacy futility monitoring

Eliminate dose j for futility when

$$\Pr(p_{E,j} < \phi_E \mid \mathcal{D}) > c_E. \tag{27}$$

A futile dose cannot receive a new cohort, serve as an IPDE destination, or enter final OBD selection. In the TITE implementation, this posterior is calculated from the selected efficacy model using the delayed-outcome likelihood.

4.3 IPDE eligibility and destination criteria

Suppose a patient was previously treated at dose ℓ and is considered for destination dose k. Recycling is permitted only when all conditions below are satisfied.

1. **Individual toxicity risk.**
$$\Pr\{q_T(\ell \to k) > \phi_T \mid \mathcal{D}\} < c_T. \tag{28}$$
   For the two complete-outcome toxicity options,
$$q_T^{(r)}(\ell \to k) = r_T + (1 - r_T)p_{T,k}, \tag{29}$$
$$q_T^{(\alpha)}(\ell \to k) = \min\{1, p_{T,k} + \alpha_T p_{T,\ell}\}. \tag{30}$$
   Under the current TITE implementation, use $q_T^{(r)}$ with $r_T = r_T^{\text{fix}}$.
2. **Expected efficacy improvement.**
$$\mathrm{E}\{q_E(\ell \to k) - p_{E,\ell} \mid \mathcal{D}\} > \delta, \tag{31}$$
   where q_E is computed under E-RD, E-AD, E-RS, or E-AS. The preferred implementation evaluates the posterior expectation draw by draw. A plug-in approximation may use posterior means of all model components.
3. **Cycle limit.** The patient satisfies $n_{\text{cycle}} < \texttt{max\_cycle}$.
4. **Completed prior-cycle outcomes.** The cycle used to establish recycling eligibility is complete, and the patient experienced neither a DLT nor an efficacy response in that completed cycle.
5. **Dose-direction restriction.** If `flexible = FALSE`, require an upward destination $k > \ell$ subject to the active dose-skipping and admissibility rules. If `flexible = TRUE`, any admissible destination may be considered.

If no destination satisfies every criterion, the patient is not recycled.

4.4 Order of operations

1. Update the toxicity posterior and the selected efficacy posterior using all currently available complete and partial information.

2. Apply dose-level overtoxicity and futility rules and update the admissible dose set.
3. Determine the next cohort dose using the active one-stage or two-stage allocation rule.
4. Evaluate each potential IPDE patient against patient- and destination-specific criteria before cohort placement.

5. Utility Functions

Utilities are based on regular-patient probabilities $p_{E,j}$ and $p_{T,j}$. Carryover parameters affect their posterior estimation through IPDE likelihood contributions, but patient-specific q_E and q_T are reserved for recycling decisions.

5.1 Utility 1: efficacy only

$$U_1(j) = p_{E,j}. \tag{32}$$

Toxicity affects the candidate set through safety rules and the MTD ceiling rather than through the numerical utility.

5.2 Utility 2: efficacy minus toxicity penalty

$$U_2(j) = p_{E,j} - \lambda_T p_{T,j}, \qquad \lambda_T \geq 0. \tag{33}$$

5.3 Utility 3: four-outcome joint score

Assign score u_{te} to joint outcome $(T = t, E = e)$. Under the working-independence approximation,

$$U_3(j) = u_{00}(1 - p_{T,j})(1 - p_{E,j}) + u_{01}(1 - p_{T,j})p_{E,j} \tag{34}$$

$$+ u_{10}p_{T,j}(1 - p_{E,j}) + u_{11}p_{T,j}p_{E,j}. \tag{35}$$

Only admissible doses are ranked using a common posterior-summary convention. A prespecified tie-breaking rule is required; selecting the lower dose is the conservative default. Utility cannot override direct toxicity-driven de-escalation.

6. Final OBD Selection

Final OBD selection is common to the one-stage and two-stage designs. The toxicity model determines the MTD ceiling, efficacy futility removes unacceptable doses, and utility ranks the remaining tried doses.

1. Select the MTD from the toxicity model. Apply the prespecified toxicity-model MTD rule. The MTD defines the upper dose ceiling for final OBD selection.

2. Remove efficacy-futile doses. Exclude every dose that satisfies the prespecified posterior futility rule.

3. Construct the OBD admissible set. C_{OBD}={j: dose j has been tried, j<=j_{MTD}, and dose j is not efficacy-futile or otherwise eliminated}.

4. Compute utility and select the OBD. Select j_{OBD}=arg $\max_{j \text{ in } C_{OBD}}$ U(j). Ties use the prespecified utility tie-breaking rule; if C_{OBD} is empty, no OBD is selected.

> **Final-selection restriction.** The response-informed allocation safeguards described in Section 3 affect interim cohort assignment only. They do not change the final OBD definition: final selection remains utility-based among tried, non-futile doses at or below the final MTD.

7. Compact Algorithms and Control Parameters

Two-stage algorithm

1. Run Stage I with toxicity-based AIDE until *s1_Max* is reached and the next toxicity recommendation is *stay*.

2. At Stage II and after every evaluated Stage II cohort, recalculate the MTD, overtoxicity, futility, observed efficacy-response indicators R_j, and A_{II}.

3. highest_utility: among admissible doses with R_j=1, assign the full cohort to the highest-utility dose; if none has a response, assign the cohort to the current MTD.

4. top2_randomized: start from the ordinary top-two utility doses. Use them if both have responses; if exactly one has a response, pair it with the MTD; if neither has a response, use the MTD and the highest admissible dose below it. Resolve duplicates with the highest distinct admissible alternative and randomize proportional to utility.

5. Continue Stage II until the prespecified stopping/sample criterion; update all model and response-informed quantities before every new Stage II cohort.

6. At final analysis, select the OBD using the common final OBD rule in Section 6.

One-stage algorithm

1. At each cohort decision, update the toxicity and efficacy models; recalculate MTD, overtoxicity, futility, response indicators R_j, and the current allocation-admissible set.

2. Compute utility for every allocation-admissible dose and define j_U as the highest-utility admissible dose.

3. If j is above the current dose, escalate toward it without skipping untried doses. If j equals the current dose, stay.

4. If j_U is below the current dose d, form A_{down}={j: j_U<=j<=d, j is toxicity-admissible/not otherwise eliminated and R_j=1}.
If A_{down} is nonempty, assign the cohort to min(A_{down}).

5. If A_{down} is empty, remain at d if d is toxicity-admissible; if d is no longer toxicity-admissible, assign the highest toxicity-admissible dose below d, subject to the global elimination rules.

6. Repeat after every evaluated cohort. Final OBD selection follows Section 6 and is not altered by the interim response-informed safeguard.

User-specified controls

Parameter	Role	Convention
`tox_model`	Complete-outcome toxicity model	`r_shared` or `alpha_shared`
`eff_model`	Efficacy carryover model	`r_dose`, `alpha_dose`, `r_shared`, or `alpha_shared`
`tite`	Partial-outcome implementation	If TRUE, toxicity uses fixed-r_T TITE; efficacy uses delayed outcome
r_T^{fix}	Fixed toxicity carryover in TITE	User specified; $r_T^{\text{fix}} = 0$ gives original TITE-CRM
τ_T, τ_E	Endpoint assessment windows	User specified
$s1\_Max$	Stage I sample-size limit	User specified
`one_stage_allocation`	One-stage cohort-allocation rule	Highest-utility allocation target; escalation does not skip untried doses
ϕ_T, c_T	Toxicity target and posterior cutoff	User specified
ϕ_E, c_E	Efficacy futility threshold and cutoff	User specified
δ	Minimum efficacy improvement for IPDE	User specified
`max_cycle`	Maximum cycles per patient	User specified
`flexible`	Permitted IPDE destination range	FALSE: upward only; TRUE: any admissible dose
`enrollment_scheme`	Cohort queue mechanism	Continuous new-patient priority or IPDE-first recruitment
`utility_type`	Dose-ranking function	1, 2, or 3
λ_T	Toxicity penalty in Utility 2	User specified
`tried`	Final-selection restriction	Required TRUE for final OBD; MTD restriction follows the toxicity model
Beta/Gamma shapes	Priors for $p_{E,j}$ and estimated carryover	Prespecified by model option; common Gamma rate for ratio representation

Stage II allocation control

stage2_allocation selects one of two response-informed Stage II rules:

top2_randomized: begin with the two admissible doses with the largest utilities. If both have at least one observed efficacy response, use them. If exactly one has a response, pair that dose with the current MTD. If neither has a response, use the current MTD and the highest admissible dose below it. Resolve duplicate doses with the highest distinct admissible alternative; if only one admissible dose remains, assign the full cohort to it. Randomization probabilities are proportional to utility.

highest_utility: among admissible doses with at least one observed efficacy response, assign the full cohort to the dose with the largest utility. If none has an observed efficacy response, assign the cohort to the current MTD.

In both versions, the MTD, overtoxicity, futility, response indicators, admissible set, and utilities are refreshed after every evaluated Stage II cohort.

A. Implementation Templates

The following JAGS-style templates show the defining structure of the six carryover options. They are schematic and omit monitoring calculations and posterior-generated quantities.

A.1 Toxicity T-R: shared r_T

```
for (j in 1:ndose) {
  p_tox[j] <- pow(skeleton[j], exp(beta))
  q_tox_r[j] <- r_t * (1 - p_tox[j]) + p_tox[j]
}
for (i in 1:N) {
  theta_tox[i] <- (1 - ipde[i]) * p_tox[dose[i]] +
                ipde[i] * q_tox_r[dose[i]]
  tox[i] ~ dbern(theta_tox[i])
}
```

For the TITE implementation, supply `r_t` as fixed data and replace the last line with `tox[i] ~ dbern(w_tox[i] * theta_tox[i])`, forcing `w_tox[i]=1` for observed DLTs.

A.2 Toxicity T-A: shared α_T

```
g_tox_carry_1 ~ dgamma(a_alpha_t, 1)
g_tox_carry_2 ~ dgamma(b_alpha_t, 1)
alpha_t <- g_tox_carry_1 / (g_tox_carry_1 + g_tox_carry_2)

for (j in 1:ndose) {
  p_tox[j] <- pow(skeleton[j], exp(beta))
}
for (i in 1:N) {
  q_tox_alpha[i] <- min(1,
      p_tox[dose[i]] + alpha_t * p_tox[previous_dose[i]])
  theta_tox[i] <- (1 - ipde[i]) * p_tox[dose[i]] +
                ipde[i] * q_tox_alpha[i]
  tox[i] ~ dbern(theta_tox[i])
}
```

A.3 E-RD: dose-specific $r_{E,j}$

```
for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
                (g_regular_1[j] + g_regular_2[j])

  g_carry_1[j] ~ dgamma(a_carry[j], 1)
  g_carry_2[j] ~ dgamma(b_carry[j], 1)
  r_e[j] <- g_carry_1[j] / (g_carry_1[j] + g_carry_2[j])
  p_ipde[j] <- r_e[j] + (1 - r_e[j]) * p_regular[j]
}
for (i in 1:N) {
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
                ipde[i] * p_ipde[dose[i]]
}
```

A.4 E-AD: dose-specific $\alpha_{E,j}$

```
for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
                (g_regular_1[j] + g_regular_2[j])

  g_alpha_1[j] ~ dgamma(a_alpha[j], 1)
  g_alpha_2[j] ~ dgamma(b_alpha[j], 1)
  alpha_e[j] <- g_alpha_1[j] / (g_alpha_1[j] + g_alpha_2[j])
}
for (i in 1:N) {
  q_eff[i] <- min(1, p_regular[dose[i]] +
      alpha_e[dose[i]] * p_regular[previous_dose[i]])
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
                ipde[i] * q_eff[i]
}
```

A.5 E-RS: shared r_E

```
g_r_1 ~ dgamma(a_r, 1)
g_r_2 ~ dgamma(b_r, 1)
r_e <- g_r_1 / (g_r_1 + g_r_2)

for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
                  (g_regular_1[j] + g_regular_2[j])
  p_ipde[j] <- r_e + (1 - r_e) * p_regular[j]
}
for (i in 1:N) {
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
                  ipde[i] * p_ipde[dose[i]]
}
```

A.6 E-AS: shared α_E

```
g_alpha_1 ~ dgamma(a_alpha, 1)
g_alpha_2 ~ dgamma(b_alpha, 1)
alpha_e <- g_alpha_1 / (g_alpha_1 + g_alpha_2)

for (j in 1:ndose) {
  g_regular_1[j] ~ dgamma(a_regular[j], 1)
  g_regular_2[j] ~ dgamma(b_regular[j], 1)
  p_regular[j] <- g_regular_1[j] /
                  (g_regular_1[j] + g_regular_2[j])
}
for (i in 1:N) {
  q_eff[i] <- min(1, p_regular[dose[i]] +
      alpha_e * p_regular[previous_dose[i]])
  theta_eff[i] <- (1 - ipde[i]) * p_regular[dose[i]] +
                  ipde[i] * q_eff[i]
}
```

A.7 Delayed-outcome efficacy likelihood in JAGS

A zeros-trick or equivalent custom likelihood is needed because the pending nonresponse contribution is $(1-\theta_{E,i})^{w_{E,i}}$ rather than a standard Bernoulli term. Define

$$\ell_{E,i} = \Delta_{E,i} Y_{E,i} \log \theta_{E,i} + \{\Delta_{E,i}(1-Y_{E,i}) + (1-\Delta_{E,i}) w_{E,i}\} \log(1-\theta_{E,i}). \tag{38}$$

Then implement $-\ell_{E,i}$ through a zeros-trick with a sufficiently large constant. Numerical protection should bound $\theta_{E,i}$ away from exactly zero and one before taking logarithms.

B. References

[1] Cheung, Y. K., and Chappell, R. (2000). Sequential designs for phase I clinical trials with late-onset toxicities. *Biometrics*, 56, 1177–1182.

[2] Lin, R., and Yuan, Y. (2020). Time-to-event model-assisted designs for dose-finding trials with delayed toxicity. *Biostatistics*, 21(4), 807–824.

[3] O'Quigley, J., Pepe, M., and Fisher, L. (1990). Continual reassessment method: a practical design for phase I clinical trials in cancer. *Biometrics*, 46, 33–48.

[4] Updated TITE-CRM Likelihoods AIDE-CRM: Updated Models and Likelihood Construction. Internal technical note, July 12, 2026.

[5] Current One-Stage and Two-Stage IPDE Design with Efficacy. Technical design specification, July–August 2026.

C. TITE Event-Clock Recycle Strategy

Scope. This section adopts the event-clock and waiting-queue strategy in the TITE-Likelihood technical note, with revised efficacy-enabled recycle eligibility. New patients are considered before eligible retreat patients. A patient becomes structurally eligible after completing the full DLT assessment without a DLT or efficacy response and may be recycled to any currently admissible dose, including the dose assigned to the regular cohort. No adjacency or one-level direction restriction is imposed. When no cohort is open, a new model-based decision remains unavailable until at least `n_eval` patients treated at the current dose have fully evaluated DLT outcomes.

C.1 Event-clock data structure

Calendar time is tracked explicitly. Each administration row records patient ID, start and evaluation times, assigned dose, cycle, cohort, and regular-versus-retreat type. At each decision attempt, toxicity and efficacy status and the applicable time-to-event weights are reconstructed from all administrations that have started.

C.2 Assignable events and waiting queue

An **assignable event** makes a treatment opportunity available for possible cohort filling:

1. a new-patient arrival; or
2. an IPDE/retreat decision at the end of a patient's assessment window. If the preceding administration completed the full DLT assessment without a DLT or efficacy response and the patient remains below the cycle limit, a retreat event is scheduled at that time.

If no cohort position is open, the event enters the waiting queue and is retained.

new	Newly arrived patient who has not yet received treatment.
retreat	Previously treated patient who completed the full DLT assessment without a DLT or efficacy response and may be available for IPDE.

The queue is ordered by event time, with sequence number breaking ties. When an open cohort position is filled, waiting new-patient events are considered first. Eligible retreat events are considered only after no waiting new patient remains. A retreat event that currently has no acceptable destination remains queued until an acceptable destination becomes available or the trial terminates.

C.3 Cohort formation and model-based decisions

The event loop is determined first by whether a cohort is already open.

Open cohort. If a cohort is open, no new regular-cohort dose decision is made. Available positions are filled from the waiting queue using new-patient-first priority. If fewer queued events are available than open positions, the cohort remains open and each later assignable event is used to fill the next position. The cohort closes only after `cohortsize` administration rows have been assigned.

No open cohort. If no cohort is open, every assignable event triggers a decision-availability check. Let $N_{\text{eval}}(d_{\text{current}})$ be the number of patients treated at the current dose whose DLT outcome is fully ascertained, either because a DLT has been observed or because the full DLT assessment window has been completed without DLT. The prespecified minimum is `n_eval`.

If $N_{\text{eval}}(d_{\text{current}}) <$ `n_eval`, the decision is unavailable: no cohort is opened, no new regular-cohort dose is assigned, and all waiting events remain queued. Each later assignable event repeats this check. The starting cohort at the prespecified initial dose is opened according to the trial initialization rule and is not blocked by this gate.

Once $N_{\text{eval}}(d_{\text{current}}) \geq$ `n_eval`, the interim toxicity and efficacy data are reconstructed, the applicable time-to-event weights are calculated, and the prespecified toxicity and efficacy models defined earlier in this document are updated. The active one-stage or two-stage design algorithm then applies its overtoxicity, futility, admissibility, no-skipping, dose-cap, suspension, and stopping rules.

If the active algorithm does not return an assignable regular-cohort dose, no cohort is opened and all waiting events remain queued. If a regular-cohort dose d_{current} is returned, a new cohort is opened and its available positions are filled immediately from the queue. A retreat patient used to fill a position is evaluated under Section C.4 and may receive d_{current} or another admissible destination.

C. TITE Event-Clock Recycle Strategy (continued)

Operational interpretation. Open cohort → fill it. No open cohort → at each assignable event, first check whether at least `n_eval` patients have fully evaluated DLT outcomes at the current dose. Threshold not met → decision unavailable and retain the queue. Threshold met → apply the prespecified models and the corresponding one-stage or two-stage algorithm. Decision returned → open a new cohort and fill it.

C.4 Retreat/IPDE eligibility and destination set

A waiting retreat event is structurally eligible for reassignment when:

1. the patient has a previous administration record;
2. the most recent administration has completed the full DLT assessment window;
3. no DLT was observed during that completed administration;
4. no efficacy response was observed during that completed administration; and
5. completed cycles are below `cycle_max`.

Structural eligibility is checked when a retreat event is considered for an open cohort position. Let $\mathcal{A}$ be the current admissible dose set produced by the most recent model update and the active one-stage or two-stage algorithm. Every $k \in \mathcal{A}$ may be considered as a recycle destination, including a lower dose, the patient's preceding dose, a higher dose, and the regular-cohort dose d_{current}. There is no requirement that $k = d_{\text{current}}$ or $k = d_{\text{last}} + 1$.

For each candidate k, calculate the IPDE toxicity and efficacy probabilities under the prespecified toxicity and efficacy model options selected earlier in this document. The patient may be assigned to k only if the model-specific IPDE toxicity-risk and expected efficacy-improvement criteria are satisfied. If no candidate satisfies these criteria, the retreat event is not used for the current position and remains queued or is removed according to the prespecified queue rule.

C.5 Operational sequence

At each assignable event:

1. update the event clock and add the new-patient or completed-cycle retreat opportunity to the queue;
2. if a cohort is open, fill its available positions from the queue using new-patient-first priority; do not make a new regular-cohort dose decision;
3. if no cohort is open, count $N_{\text{eval}}(d_{\text{current}})$, the patients at the current dose with fully ascertained DLT outcomes;
4. if $N_{\text{eval}}(d_{\text{current}}) <$ `n_eval`, declare the decision unavailable, keep the cohort closed, retain the waiting queue, and wait for the next assignable event;
5. if $N_{\text{eval}}(d_{\text{current}}) \geq$ `n_eval`, reconstruct the interim data, calculate the applicable time-to-event weights, update the selected toxicity and efficacy models, and apply the active one-stage or two-stage algorithm;
6. if the algorithm returns no assignable dose, keep the cohort closed and retain the waiting queue;
7. if the algorithm returns d_{current}, open a new cohort and fill it from the queue; for each retreat event considered, evaluate all currently admissible destinations under Section C.4; and
8. continue until the administration cap or another stopping condition is reached.

C.6 Relationship to efficacy-based IPDE criteria

The event-clock rules determine *when* a patient becomes available and when a cohort decision is attempted. The selected toxicity and efficacy models determine the patient-specific IPDE risk and efficacy-improvement quantities. The `n_eval` gate first determines whether a model-based regular-cohort decision may be attempted. Once that gate is met, the active one-stage or two-stage algorithm determines whether an assignable regular-cohort dose is returned and defines the current admissible dose set. Queue priority determines only which waiting event is considered first; it does not determine the dose.

Implementation rule. Any currently admissible dose may be selected for an eligible retreat patient, including the regular-cohort dose. New-patient-first priority does not override destination-specific toxicity, futility, or efficacy-improvement criteria.

C.7 Queue behavior and edge cases

- If one or more new patients are waiting, the earliest waiting new patient is assigned before any retreat event.
- A retreat event with no currently acceptable destination is not automatically discarded; it may remain queued until an acceptable destination becomes available or the trial ends.
- If a cohort remains partially filled, later assignable events fill the open positions without a new dose decision.
- When no cohort is open and fewer than `n_eval` patients have fully evaluated DLT outcomes at the current dose, each assignable event leaves the decision unavailable and the queue unchanged except for the newly added event.
- The administration cap counts both regular and retreat administrations.
- A patient can generate another retreat event only after the latest administration completes the full DLT assessment without a DLT or efficacy response and the cycle limit has not been reached.

C.8 Required reporting

Report queue entries and waiting times by event type; the value of $N_{\text{eval}}(d_{\text{current}})$ at each closed-cohort assignable event; events at which the `n_eval` gate prevents a decision; model-based decision attempts with and without an assignable regular-cohort dose; retreat eligibility and reasons; preceding and destination doses; assignments to d_{current} versus other admissible doses; events queued at termination; and total regular plus retreat administrations.